

Sciatic Nerve

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1. Definition

- **Sciatic nerve** is the **thickest nerve in the body**.
- It is the **largest branch of the sacral plexus**.
- It terminates by dividing into:
 - **Tibial nerve**
 - **Common peroneal nerve**

2. Origin & Root Value

- **Root value: L4, L5, S1, S2, S3**
- It has two parts:

Part	Root contribution
Tibial part	Ventral divisions of L4, L5, S1, S2, S3

Part	Root contribution
Common peroneal part	Dorsal divisions of L4, L5, S1, S2

Easy idea

Sciatic = L4 → S3

Then remember:

Tibial = Ventral

Common peroneal = Dorsal

3. Course

Think of the sciatic nerve as simply:

Pelvis → below piriformis → between GT & IT → thigh → popliteal fossa → splits

A. In pelvis

- Lies **in front of piriformis**.
- Covered by **fascia of piriformis**.

B. In gluteal region

- Enters through the **greater sciatic foramen**.
- Passes **below piriformis**.
- Runs downward with slight **lateral convexity**.
- Passes between:
 - **Ischial tuberosity medially**
 - **Greater trochanter laterally**

Most important picture in your head

Piriformis

↓

Sciatic nerve comes BELOW it



Ischial tuberosity ← nerve → **Greater trochanter**

C. In thigh

- Enters posterior thigh at the **lower border of gluteus maximus**.
- Runs **vertically downward**.
- Reaches the **superior angle of popliteal fossa**.
- Divides into:
 - **Tibial nerve**
 - **Common peroneal nerve**

📌 **Division:** approximately at the junction of **upper 2/3** and **lower 1/3** of thigh.

4. Relations

A. Gluteal region

Don't memorize "superficial/deep/medial" randomly. Think:

Behind

- **Gluteus maximus**

In front

1. Body of ischium
2. Obturator internus + gemelli
3. Quadratus femoris
4. Obturator externus
5. Capsule of hip joint
6. Upper transverse fibres of adductor magnus

Medial

- **Inferior gluteal nerve**

- Inferior gluteal vessels

One-line memory

Sciatic nerve is sandwiched:

Gluteus maximus behind + deep muscles/hip joint in front

B. Relations in Thigh

This is much easier.

Direction	Relation
Posterior / superficial	Long head of biceps femoris crosses it
Anterior / deep	Adductor magnus
Medial	Semimembranosus + semitendinosus
Lateral	Biceps femoris

Visualize the nerve in the middle:

Semimembranosus + Semitendinosus  **SCIATIC NERVE**  Biceps femoris

And **Adductor magnus** is underneath/deep to it.

5. Companion Artery

- Sciatic nerve is accompanied by a small artery called **arteria nervi ischiadici**.
- It is a branch of the **inferior gluteal artery**.
- It accompanies the nerve for some distance before entering its substance.

6. Branches

A. Articular branches

- Supply the **hip joint**.
- Arise in the **gluteal region**.

B. Muscular branches

Tibial part supplies:

- Semitendinosus
- Semimembranosus
- Long head of biceps femoris
- Hamstring/ischial part of adductor magnus

Common peroneal part supplies:

- Short head of biceps femoris **ONLY**

High-yield

Sciatic → mostly hamstrings through tibial part; only short head of biceps through common peroneal part.

C. Terminal branches

- Tibial nerve
- Common peroneal nerve

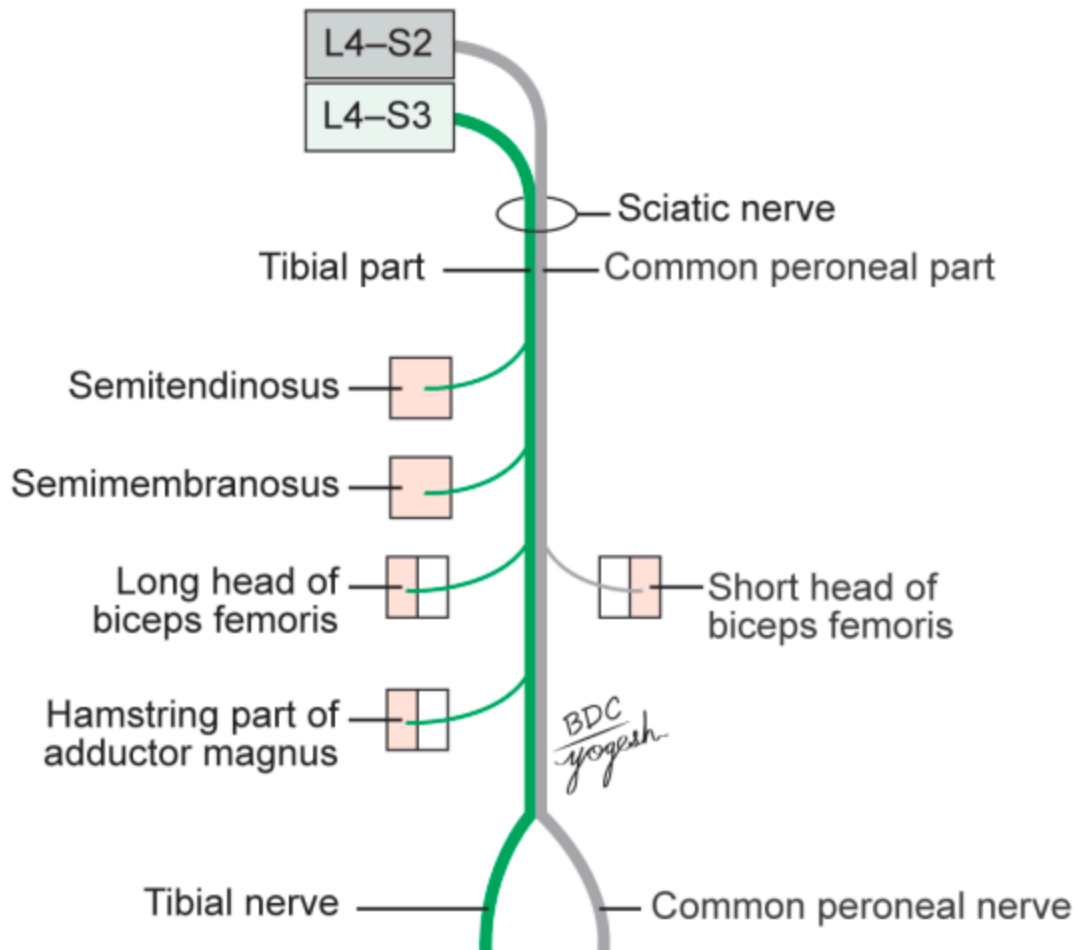


Fig. 7.5: Branches of sciatic nerve

7. Clinical Anatomy of Sciatic Nerve

1. "Sleeping foot"

- When sitting on a **hard edge**, the sciatic nerve may be compressed between the **edge of the chair/table and femur**.
- Causes **temporary numbness/tingling of the lower limb**.
- Sensation returns after relieving the pressure → commonly called "**sleeping foot**."

2. Sciatica ★

- Shooting pain along the distribution of the sciatic nerve and its terminal branches.

- Usually begins in the **gluteal region** and radiates:
Back of thigh → lateral side of leg → dorsum of foot
- Usually caused by **compression of one or more nerve roots** forming the sciatic nerve.
- Common causes:
 - **Intervertebral disc prolapse**
 - Neuritis

3. Sciatic nerve injury ★

Causes:

- Penetrating wounds
- Hip dislocation

Motor loss:

- Loss of **hamstring function**
- **Foot drop**
- Loss of:
 - Dorsiflexion
 - Plantar flexion
 - Eversion
 - Movements of muscles of the sole

Sensory loss:

- Back of thigh
- Whole leg
- Foot

Exception: Area supplied by the **saphenous nerve** is spared.

★ Exam-ready summary

Sciatica = pain along sciatic distribution.

Sciatic injury = motor + sensory loss, with **foot drop** and sparing of the **saphenous nerve** area.

★ Ultra-High-Yield Revision

If you have only 1 minute before the exam:

Sciatic nerve

- Thickest nerve
- Largest branch of sacral plexus
- **L4–S3**
- **Below piriformis**
- Between **ischial tuberosity & greater trochanter**
- Descends posterior thigh
- Divides at **upper 2/3 + lower 1/3 junction**
- **Tibial + common peroneal**

Relations

Gluteal:

Behind: Gluteus maximus

In front: deep structures + hip joint

Medial: inferior gluteal nerve/vessels

Thigh:

Medial: Semimembranosus + Semitendinosus

Lateral: Biceps femoris

Deep: Adductor magnus

Crossed superficially by: Long head of biceps femoris